

NIKHIL CHAUDHARY

Unity Game Developer - Multiplayer & Mobile Specialist

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PROFESSIONAL SUMMARY

Unity Game Developer with over 2.6 years of production experience developing Android, iOS, WebGL, and Windows games using Unity (2021 LTS, 2022 LTS, 6 LTS) and C#. Specialized in multiplayer game development, scalable gameplay architecture, backend integration, mobile optimization, and production-ready gameplay systems. Experience contributing to live production titles with 5M+ downloads across casual, hyper-casual, and multiplayer games.

Strong expertise in Photon PUN2, Photon Fusion, Unity NGO, Socket.IO, Colyseus, Firebase, Node.js, MongoDB, URP, gameplay systems, clean architecture, object pooling, responsive UI systems, and real-time networking workflows. Experienced in feature ownership, live production support, Agile/sprint delivery, code reviews, reusable system design, multiplayer synchronization, performance optimization, and scalable maintainable code architecture.

Delivered a 1900% export pipeline optimization improving workflow scalability from 20 exports to 1000+ exports in 5 minutes. Immediate joiner.

ACHIEVEMENTS & RECOGNITION

- **Ranked 1st** in Deftsoft internal Unity challenge and game-development quiz - evaluated on clean architecture, gameplay engineering, and technical problem-solving across the full engineering team.
- **1900% export pipeline optimization** - redesigned a production pipeline from 20 exports to 1000+ exports in 5 minutes using coroutine-based processing and memory optimization, directly unblocking production team workflow.
- **Contributed to a live title with 5M+ downloads** - shipped features and optimizations on a production mobile game (Alnahsha Run) reaching 5 million downloads across Android and iOS.
- **Mentored junior developers, improving team delivery by 12%** - created reusable UI tooling that enabled a junior developer to complete a delivery 1.5 days ahead of schedule, with measurable improvement in overall sprint velocity.

TECHNICAL SKILLS

Game Engines: Unity 2D/3D (Advanced, Unity 2021 LTS / Unity 2022 LTS / Unity 6 LTS), Unreal Engine (Beginner), VR/AR/XR Prototype Development

Render Pipelines: Universal Render Pipeline (URP), High Definition Render Pipeline (HDRP), Built-in Render Pipeline, Scriptable Render Pipeline (SRP), Shader Graph, Post-Processing Stack, Lighting and Rendering Optimization, GPU Instancing, Occlusion Culling, Mobile Rendering Optimization, WebGL Rendering

Programming Languages: C#, C++, C, JavaScript

Multiplayer and Networking: Photon PUN2, Photon Fusion 2.x, Unity NGO (Netcode for GameObjects), Socket.IO, Colyseus, Mirror (awareness), FishNet (awareness), Unity Relay, Unity Lobby, Dedicated Server Architecture, Server-Authoritative Multiplayer, Matchmaking Systems, Lobby Systems, Room Management, Player Synchronization, RaiseEvent Synchronization, RPC Systems (ClientRpc / ServerRpc), Network Object Spawning, NetworkTransform, NetworkVariable, Authority Handling, Client-Side Prediction, Lag Compensation, Reconnection Systems, Network Optimization, Multiplayer UI Flows, Social Invite Systems, High CCU Optimization

Backend and Live Services: Firebase Authentication, Firestore, Realtime Database, Push Notifications, Node.js, Express.js, MongoDB, REST APIs, JSON Serialization, Cloud Save Systems, PlayFab (awareness), Live-Service Connectivity, Backend Synchronization

Gameplay Systems: Gameplay Architecture, Combat Systems, Health Systems, Resource Management, Building Systems, Endless Runner Systems, Tilemap Systems, State-Driven Gameplay, Wave Systems, Save/Load Systems, Enemy AI Systems, FSM AI, Behavior Trees (awareness), Progression Systems

Architecture and Design Patterns: Clean Architecture, Modular Architecture, SOLID Principles, MVP, MVC, Singleton, State Machine, Factory, Strategy, Observer, Event-Driven Architecture, ScriptableObject-Driven Architecture, Object Pooling, Reusable Gameplay Frameworks

Performance Optimization: Unity Profiler, Memory Optimization, GC Optimization, Draw Call Reduction, FPS Optimization, Mobile Optimization, WebGL Optimization, Build Size Optimization, Low-End Device Optimization (Snapdragon 4xx, Mali GPU), Texture Optimization, Sprite Atlas Optimization, IL2CPP Build Target

Mobile Development: Android and iOS Deployment, APK/AAB Generation, Play Store / App Store Submission, App Signing, Touch Controls, Resolution Handling, Safe Area Handling, Battery Optimization, Cross-Platform Development

CI/CD and Tools: Unity Cloud Build, GitHub Actions (Unity CI/CD), Git, GitHub, GitLab, Sourcetree, Visual Studio, Android Logcat, DOTween, Unity Addressables, AdMob, IronSource, AppLovin, Firebase Analytics, Jira, Trello, ClickUp, Agile / Scrum / Sprint Delivery

UI and UX: Unity UI, TextMeshPro, Responsive UI Scaling, Localization-Ready UI Systems, HUD Systems, UI Animation Polish, Cross-Device UI, Figma

PROFESSIONAL EXPERIENCE

Unity Game Developer

Deftsoft

Oct 2023 - Feb 2026

Mohali, Punjab

- Developed and maintained multiplayer, casual, and hyper-casual games across Android, iOS, WebGL, and Windows platforms using **Unity 2021/2022 LTS/6 LTS** and C#, contributing to live production titles with **5M+ downloads**.
- Built real-time multiplayer gameplay systems using **Photon PUN2**, **Photon Fusion 2.x**, **Unity NGO**, and **Socket.IO**, including dedicated server architecture, matchmaking, room systems, player synchronization, leaderboard systems, friend systems, reconnection flows, and multiplayer UI flows.
- Integrated backend systems using **Firestore**, **Node.js**, **Express.js**, and **MongoDB** for authentication, real-time game-play synchronization, persistent progression systems, cloud saves, push notifications, and live-service connectivity.
- Optimized a production export pipeline from approximately **20 exports to 1000+ exports within 5 minutes**, achieving a **1900% performance improvement** through coroutine-based processing, memory optimization, and workflow restructuring.
- Developed scalable gameplay systems including object pooling, FSM-based enemy AI, wave systems, gameplay state handling, responsive URP-compatible UI systems, resource systems, building systems, and reusable gameplay frameworks.
- Designed clean, maintainable, and modular architecture using SOLID principles, ScriptableObject workflows, event-driven systems, FSM-based gameplay, and scalable project structures; participated in code reviews and pull request workflows.
- Implemented performance optimization for Android/iOS low-end devices (Snapdragon 4xx, Mali GPU) including GC optimization, draw call reduction, WebGL optimization, IL2CPP build target configuration, and mobile profiling using Unity Profiler.
- Created reusable UI tooling for internal team workflows, enabling a junior developer to complete delivery **1.5 days ahead of schedule** while improving overall team efficiency by **12%**.
- Collaborated with designers, artists, QA, and engineers in **Agile/sprint** production workflows, contributing to technical design documentation, feature ownership, production maintenance, and live production support.
- Mentored junior developers on Unity workflows, multiplayer systems, clean architecture, debugging strategies, and maintainable gameplay engineering.

KEY PROJECTS

Alnahsha Run - Unity 2021 LTS, C#, URP, Android/iOS

5M+ Downloads

- Endless runner game across Riyadh, Dubai, and Cairo environments - shipped to 5M+ downloads on Android and iOS, with stable **60 FPS performance on low-end devices** (Snapdragon 4xx / Mali GPU class).
- Implemented URP-compatible mobile optimization, dynamic UI systems, gameplay balancing, lightweight architecture, responsive UI scaling, and low-end device optimization.

Multiplayer Card Game - Unity, Photon PUN2, Socket.IO, Firestore, Android/iOS

Production

- Developed poker-style real-time multiplayer gameplay with **server-authoritative matchmaking**, lobby systems, friend systems, push notifications, leaderboard systems, and player synchronization across concurrent sessions.
- Integrated Firestore authentication, WhatsApp social invite systems, localization-ready multiplayer UI, backend synchronization, and persistent online progression systems.

Multiplayer Arena Protocol - Unity 2022 LTS, C#, Netcode for GameObjects, Windows

2-Player Co-op

- Built a server-authoritative 2-player co-op survival arena using **Unity Netcode for GameObjects (NGO)**, featuring a modular ability system (Dash, Projectile, Heal) with independently networked cooldowns.
- Implemented a server-only enemy AI state machine (Patrol - Chase - Attack) synchronized across clients via **NetworkTransform** and **NetworkVariable**.
- Designed a slot-based session reconnection system that restores player health, score, and ability cooldowns on rejoin, surviving NGO clientId reassignment across reconnects.

City Builder Resource Management - Unity 6 LTS, Node.js, MongoDB, WebGL + Android

Confidential

- Developed scalable resource management and building systems with clean modular gameplay architecture, object pooling, and optimized gameplay synchronization.
- Integrated Node.js backend, MongoDB database workflows, and production-ready gameplay maintenance; optimized for WebGL delivery with draw call reduction and build size optimization.

3D Character Creator Tool - Unity 2022 LTS, C#, Blendshapes, Windows

Confidential

- Built advanced character customization including facial morphing, blendshape systems, color customization, and export workflows for a Windows production tool.
- Resolved the production export pipeline bottleneck that became the **1900% optimization** achievement, scaling from 20 exports to 1000+ exports within 5 minutes.

Multiplayer Tank Battle - *Unity 2022 LTS, Photon PUN2 to Fusion 2.x, Android*

In Progress

- Developed a **Photon PUN2** multiplayer build featuring offline AI, online 1-4 player rooms, **RaiseEvent** synchronization, and synced scene loading using **Photon Custom Properties**.
- Currently migrating the networking layer from **Photon PUN2** to **Photon Fusion 2.x Host Mode** for client-side prediction, **Unity Relay** integration, and improved network synchronization.
- Built ScriptableObject-driven tank systems, FSM-based enemy AI, matchmaking, and responsive multiplayer UI, carrying the gameplay systems across the networking migration.

PORTFOLIO & PUBLISHED PROJECTS

Portfolio: nikhilchaudhary285.github.io

Published Game: Google Play, App Store

Demo Videos: Alnahsha Run, Multiplayer Card Game, Multiplayer Tank Battle

EDUCATION

Bachelor of Computer Applications (BCA)

2020 - 2023

Guru Nanak Dev University (GNDU)

Amritsar, Punjab

Relevant coursework: Data Structures, Object-Oriented Programming, Database Management Systems, Software Engineering, Computer Networks, Operating Systems.

CERTIFICATIONS

Unity Junior Programmer Pathway - Unity Technologies

credly.com/go/oMRb0H0K

Formal Credly-verified validation of core Unity/C# fundamentals, completed alongside ongoing production work.

CORE TECHNOLOGIES

Unity Game Developer, Multiplayer Unity Developer, Unity 2D/3D, Unity 2021 LTS, Unity 2022 LTS, Unity 6, URP, Photon PUN2, Photon Fusion, Unity NGO, Socket.IO, Colyseus, Mirror, Firebase, Node.js, MongoDB, Multiplayer Architecture, Dedicated Server Architecture, Server-Authoritative Multiplayer, Real-Time Networking, Client-Side Prediction, Lag Compensation, Unity Relay, Unity Lobby, Mobile Game Development, Android Game Development, iOS Game Development, Hyper-Casual Games, Casual Games, Multiplayer Games, Gameplay Systems, ScriptableObjects, State Machines, Behavior Trees, Event-Driven Architecture, Object Pooling, Backend Integration, WebGL Optimization, Mobile Optimization, IL2CPP, Production Stability, Cross-Platform Development, Android and iOS Deployment, APK/AAB Pipeline, Agile, Scrum, Sprint Delivery, Code Reviews, Responsive UI Systems, Localization Systems, Production Game Development, Live Ops Support, Performance Optimization, Reusable Gameplay Systems, Clean Architecture, Scalable Architecture, SOLID Principles, Unity Junior Programmer Pathway Certified.